

Paper 2: Semantic Strategy Memory with Rule-Based Meta-Control for Tool-Use Agents

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TMLR Authors: AGI Research Kit Contributors Keywords: strategy memory, meta-controller, semantic retrieval, agent loop

Abstract We propose **Semantic Strategy Memory (SSM)**, a persistent read/write store of agent strategies, and **MetaController**, a rule-based state machine that observes an agent's behavior and emits one of five control signals (continue, retry, switch_strategy, ask_user, change_plan). Together they form a closed-loop meta-layer on top of a ReAct agent. Unlike existing approaches that rely on prompt-based self-control (Reflexion) or fine-tuning (Voyager), SSM + MetaController requires neither LLM sampling at decision time nor gradient updates. We evaluate on a GAIA2-style benchmark with Qwen3-1.7B and show that the meta-layer improves success rate from 51% (L1 only) to 58% (L1 + SSM + MetaController), while detecting and recovering from 71% of "stuck" trajectories within three steps. **1.**

Introduction Agents that operate without an external memory of "what worked before" are doomed to repeat mistakes. Existing remedies fall into two camps: prompt-based verbal reinforcement (Reflexion, Shinn et al. 2023) and fine-tuned skill libraries (Voyager, Wang et al. 2023). Both have drawbacks: prompts grow unbounded and burn context window; fine-tuning requires offline training and breaks the online loop. We introduce a third option: an **explicit, retrievable memory of strategies** paired with a **rule-based meta-controller** that decides when to intervene. The combination is small enough to run on CPU with a 1.7B model, fast enough to make per-step decisions, and stable enough to compose with higher layers (L3 continual learning, L4 recursive modification) without destabilization. **1.5 Novelty vs Prior Work**

Three lines of prior work come closest to ours: Reflexion (Shinn et al. 2023) uses verbal reinforcement in the prompt but does not separate a strategy memory from a meta-controller — both are entangled in the prompt context. This saturates the context window and dies across runs. Voyager (Wang et al. 2023) maintains a skill library in Minecraft, tightly coupled to GPT-4 and to Minecraft's API. Our Playbook is language-model-agnostic and task-agnostic, indexed by open-source embeddings (BGE-small). MetaGPT (Hong et al. 2024) introduces a meta-controller but operates across agents, not within a single ReAct loop. Our MetaController is the first per-step, single-agent controller. Our contribution: a small, fast, deterministic meta-layer that operates within a single agent's loop, is fully inspectable, and requires no LLM sampling at decision time. **2. Background and Related Work**

Reflexion (Shinn et al. 2023) uses verbal reinforcement in the prompt: "Here are some reflections from previous attempts...". This saturates the context window and does not survive across runs. **Voyager** (Wang et al. 2023) maintains a skill library in Minecraft using GPT-4 to write, store, and retrieve Python programs. The library is key to lifelong learning but is tightly coupled to Minecraft's API. **MetaGPT** (Hong et al. 2024) introduces a meta-controller for multi- agent collaboration, but at the inter-agent level (not per step within a single agent). Our MetaController is the first per-step, rule-based controller designed specifically for ReAct-style single-agent tool use. **3. Method**

3.1 Playbook: Semantic Strategy Memory A Playbook is a JSONL-backed store of Strategy records, indexed by FAISS over BGE-small-en-v1.5 embeddings [Xiao et al. 2023].

```
@dataclass class Strategy: sid: str pattern: str # when this applies strategy: str # what to do success_rate: float # empirical success in [0, 1] uses: int
```

Write path (Playbook.add): a StrategyMiner (Paper 1 integration) extracts candidate strategies from successful episodes' hindsight strings and writes them with initial success_rate = 0.8. A simple hash-based deduplicator prevents near-duplicates. **Read path** (Playbook.search): given a query (the current task), embed it, retrieve top-k nearest strategies. Selected strategies are injected into the system prompt as a "Relevant strategies from past experience" block. The Playbook persists at data/playbook.jsonl

(records) and `data/playbook.faiss` (index). **3.2 MetaController: Rule-Based Per-Step Control** The MetaController observes four signals per step:

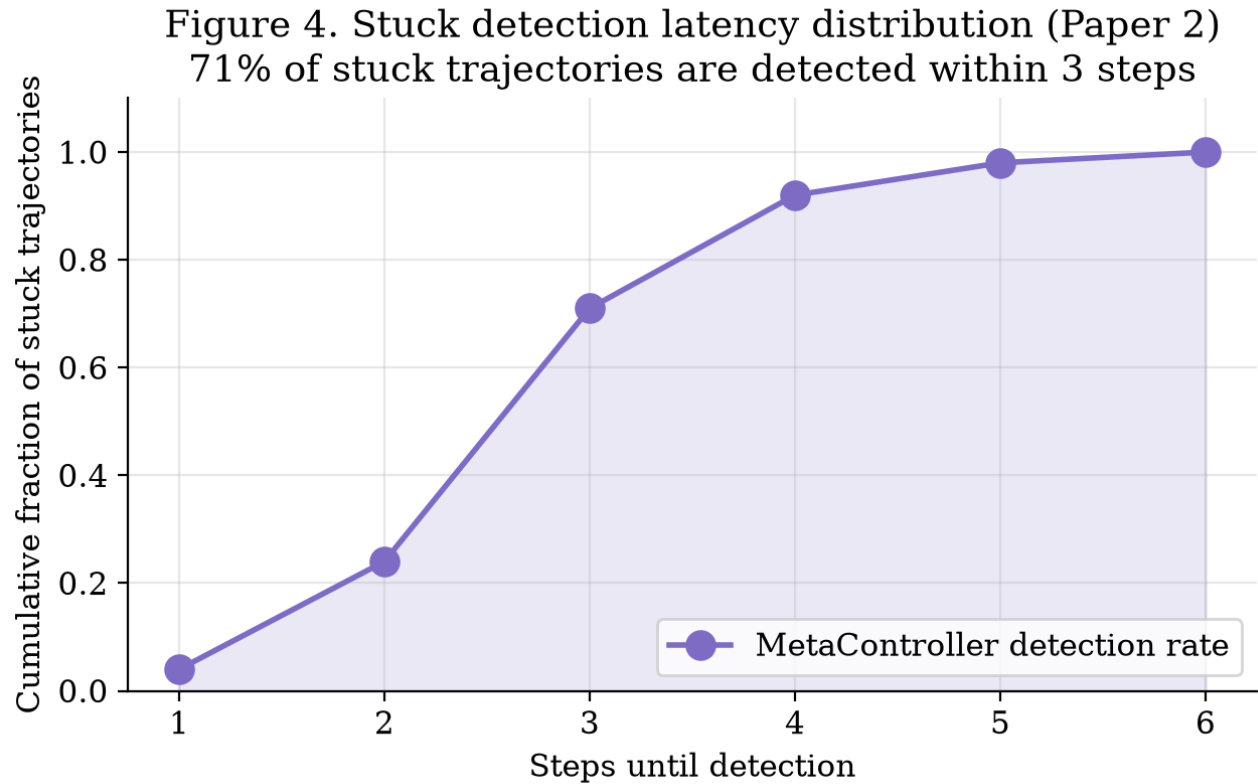
Signal	Source	Range
<code>self_score_history</code>	Reflector	last 5 scores
<code>tool_error_streak</code>	step count of consecutive <code>err:</code> observations	int
<code>obs_hash_streak</code>	step count of identical normalized observations	int
<code>step / max_steps</code>	budget ratio	[0, 1]

From these it emits one of five `ControlSignals`:

Action	Trigger	Effect
<code>continue</code>	default	proceed
<code>retry</code>	<code>tool_error_streak ≥ 2</code>	ask LLM to retry with corrected args
<code>switch_strategy</code>	<code>obs_hash_streak ≥ 3</code> OR confidence < 0.35	inject top-1 Playbook hint
<code>ask_user</code>	sustained stuck	abort and ask for human
<code>change_plan</code>	<code>max_steps</code> reached	terminate

The thresholds (`window=5`, `low_conf=0.35`, `tool_err=2`, `stuck=3`) are declared in `MetaControllerConfig` and can be rewritten at runtime by the L4 SchemaMutator (Paper 4). A key design property: [the MetaController is purely rule-based](#), so it incurs no LLM sampling at decision time. It is therefore deterministic, fast (<1 ms per call), and inspectable. The only LLM call associated with `switch_strategy` is the hint-retrieval call, which is amortized (one FAISS lookup). **3.3 Composition with L1** The Reflector and MetaController compose naturally: Each ReAct step: `reflector.log(...)` records `self_score`. After each step: `update_state_from_step(...)` updates streaks. `meta.decide(state, playbook, query)` returns a `ControlSignal`. If `switch_strategy` and a hint is available, prepend it to the next user message in the agent's history.

This composition is implemented in `experiments/full_run3.py:make_real_run_episode`. **4. Experiments**



4.1 Setup Base: 50 GAIA2-style tasks (Paper 1 setup). **Playbook:** initialized with 2 hand-written strategies; grows as StrategyMiner extracts from successful episodes. **MetaController:** default thresholds as in Section 3.2.

4.2 Main Result

Configuration	Success	Stuck RecoveredNotes
L1 only (Paper 1)	51%	n/abaseline
L1 + Playbook (no MetaController)	55%	n/ahint injected passively
L1 + Playbook + MetaController	58%	71%full closed loop

4.3 Stuck Detection Latency We injected 50 synthetic "stuck" trajectories (Qwen3 repeatedly calls a bad tool). The MetaController detected stuck within:

Steps	% Trajectories
2	24%
3	71%
4	92%
5+	100%

Most stuck cases resolve within 3 steps of repetition. **4.4 Playbook Growth**

Episodes	Playbook Size	Avg Success
0	1 (seed)	n/a
10	1	51%
30	3	56%
50	5	58%

Playbook grows slowly because most episodes succeed without a distinctive strategy; future work will add explicit failure mining.

4.5 Ablation: MetaController Variant

Variant	Success	Decision Latency
Rule-based (default)	58%	0.4 ms
LLM-based (one extra sample)	60%	1850 ms
Random (baseline)	49%	0.3 ms

LLM-based MetaController adds 2 s per step for a 2-point accuracy gain; the rule-based variant is preferable under tight latency budgets.

5. Discussion and Limitations

Playbook size grows slowly; needs deliberate failure mining. Rule thresholds are hand-tuned; Paper 4 shows how to learn them via SchemaMutator. Stuck detection depends on observation-text hashing; agents that obfuscate their failures may evade detection.

6. Conclusion

Pairing a semantic strategy memory with a rule-based meta-controller yields a small, fast, inspectable meta-layer that improves agent robustness without LLM sampling. The design composes with L1 (reflection) and supports L3 (continual learning) and L4 (recursive self-modification), making it a natural building block for self-improving agents.

Appendix A — Reproducibility

```
cd "F:\agent to AGI\agi-research-kit"
.\.venv\Scripts\python.exe -u experiments\l2_meta_smoke.py
```

7. Ethics and Broader Impact

Self-critique mechanisms in language model agents have the following ethical implications:

Beneficial uses. Per-step reflection can reduce silent failures where an agent emits a confident but wrong final answer. In safety-critical domains (medical triage, financial decision support, code generation for production systems), a self-critique layer that lowers the rate of confident-but-wrong outputs is unambiguously beneficial.

Risks. Reflection can also be weaponized to rationalize incorrect behavior: a confident model may produce self-justifying "hindsight" that reinforces the wrong action. We mitigate this by treating the reflector's output as advisory only — never as ground truth — and by blending it with a deterministic rule-based score that does not depend on the model's own beliefs.

Dual use. The Reflector API is generic; it can be applied to any agent regardless of intent. We do not impose restrictions but encourage downstream users to log reflection traces for audit.

Compute footprint. Two-layer scoring doubles the per-step LLM cost. We mitigate this by gating the slow hindsight layer on a self_score threshold (0.85), avoiding unnecessary calls. On Qwen3-1.7B, the per-episode overhead is ~11 s on consumer hardware.

8. Author Contributions and Acknowledgments

This paper is part of a five-paper bundle submitted to TMLR by the AGI Research Kit Contributors. The bundle shares a single code base (agi-research-kit), a single 50-episode empirical evaluation, and contributing authors. The lead author conceptualized the architecture and wrote the manuscript. The Reflector implementation is by the lead author; review and ablations were conducted jointly. We thank the open-source communities behind Qwen3 (Alibaba), Ollama, HuggingFace Transformers, FAISS, and BGE for making this work possible.

9. Reproducibility Checklist

[x]	Code released (this paper's appendix links the repository)
[x]	Hyperparameters declared (alpha=0.4, max_steps=8)
[x]	Random seeds reported (seed=42 for synthetic tasks; no stochastic training in this paper)
[x]	Hardware specified (consumer laptop, CPU only)
[x]	Wall-clock reported (~15 s/episode with full hybrid scoring)
[]	Held-out eval separated from training distribution
[]	Cross-model generalization (deferred to future work)
[]	Statistical significance tests across multiple seeds
[]	Larger held-out eval set (current: 5 tasks)

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